SDITO LAB

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Assignment 07

Problem Statement : Hosting a website on EC2

Aim

To launch an Amazon EC2 instance, configure a web server, and host a static website using terminal-based configuration.

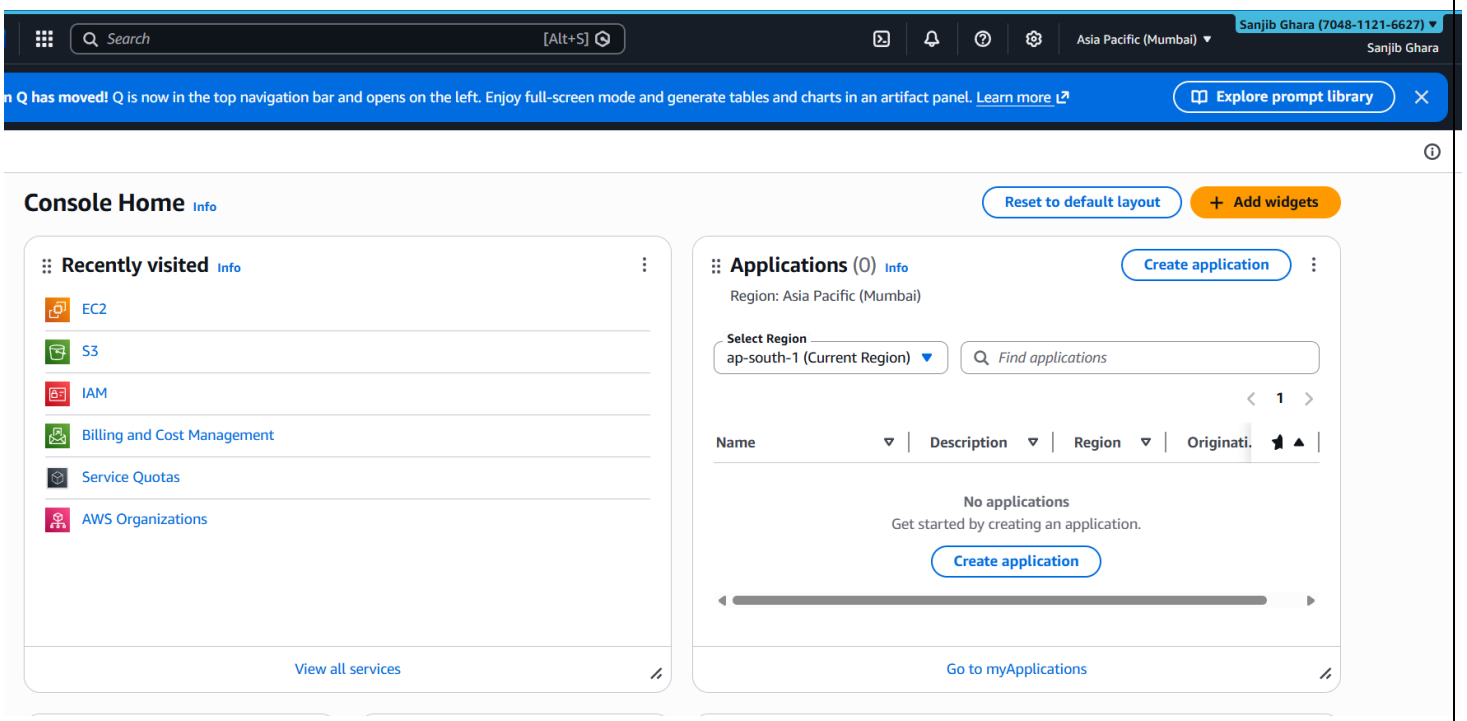
Requirements

- AWS Account
- Key pair (.pem file)
- HTML file (index.html)
- Internet connection
- Bitvise SSH Client

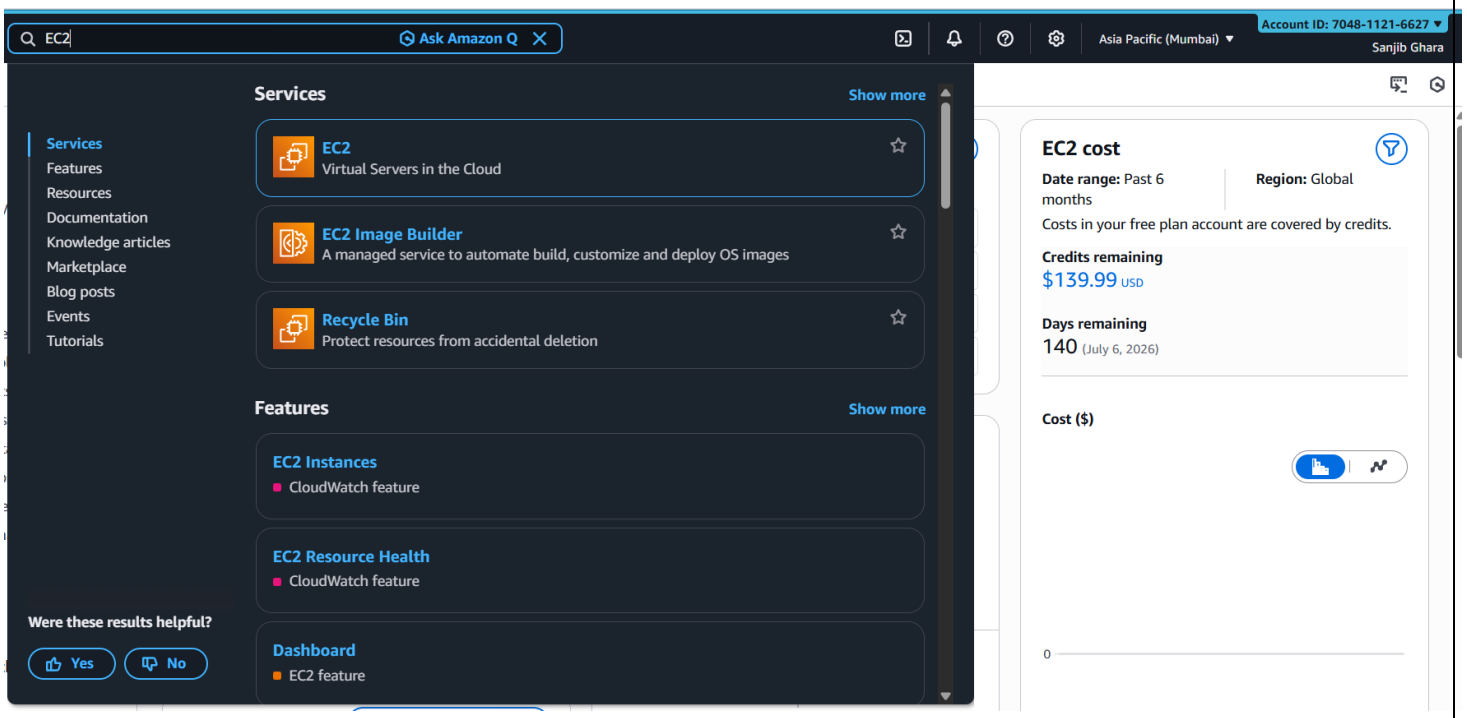
Solution:

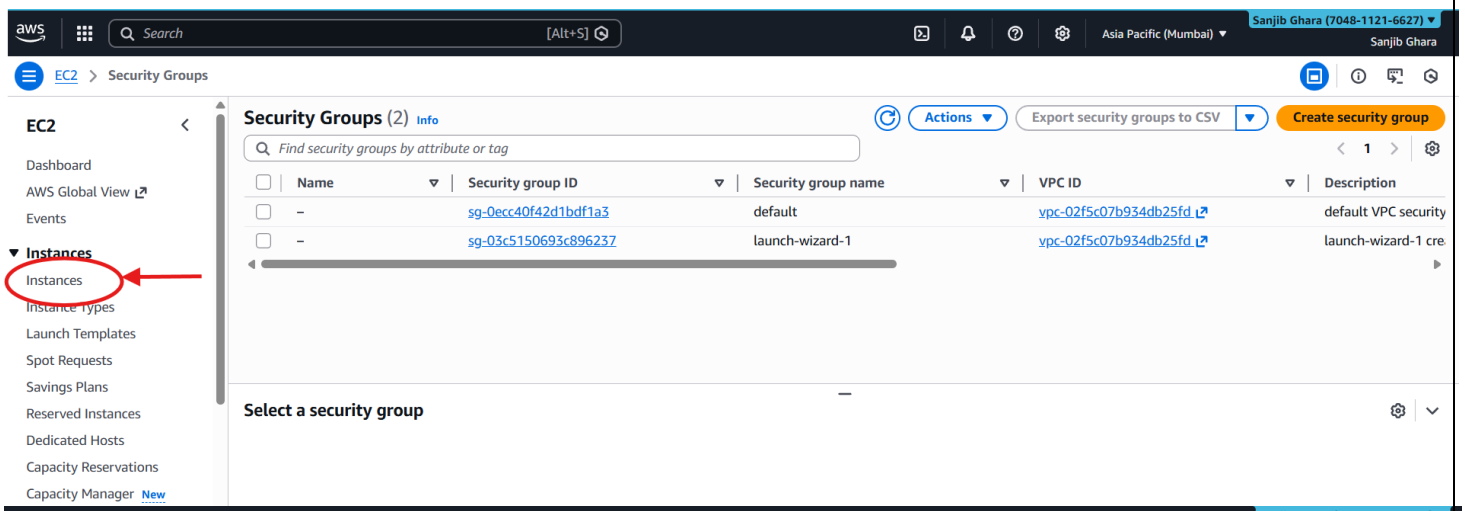
Step 1: Launch an EC2 Instance

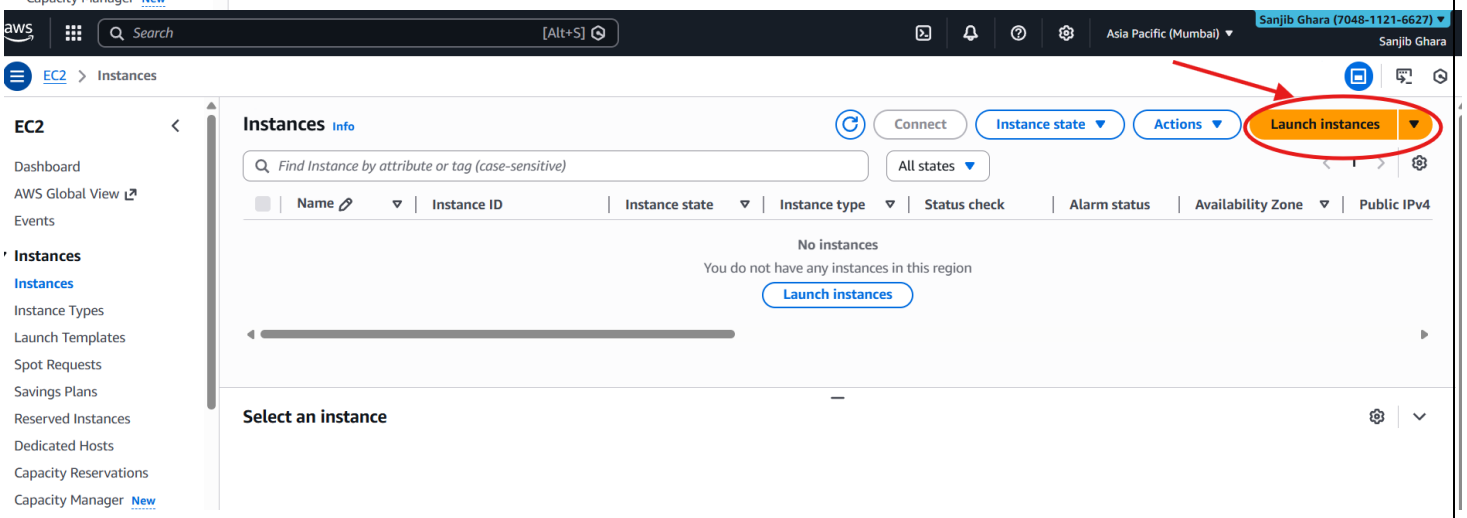
1. Login to AWS Management Console.



2. Search for EC2 and open the service.



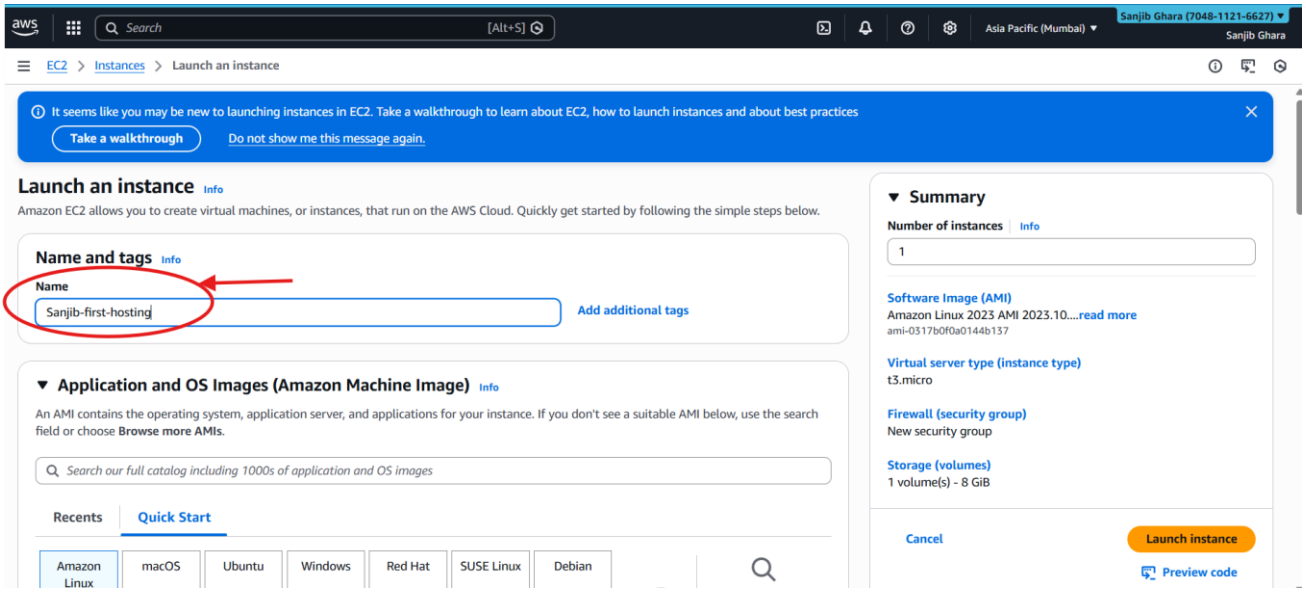


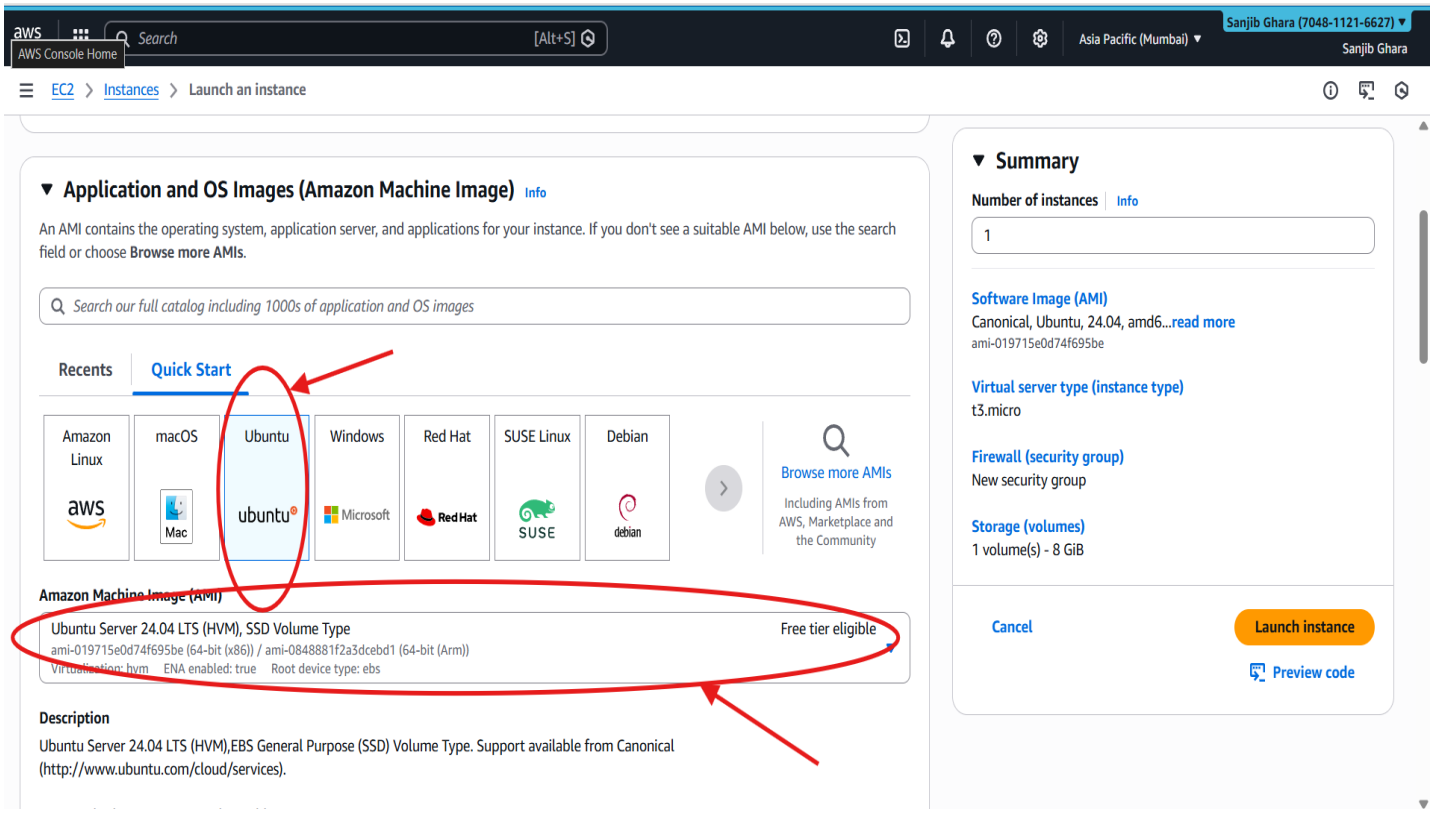


3. Click Launch instance.

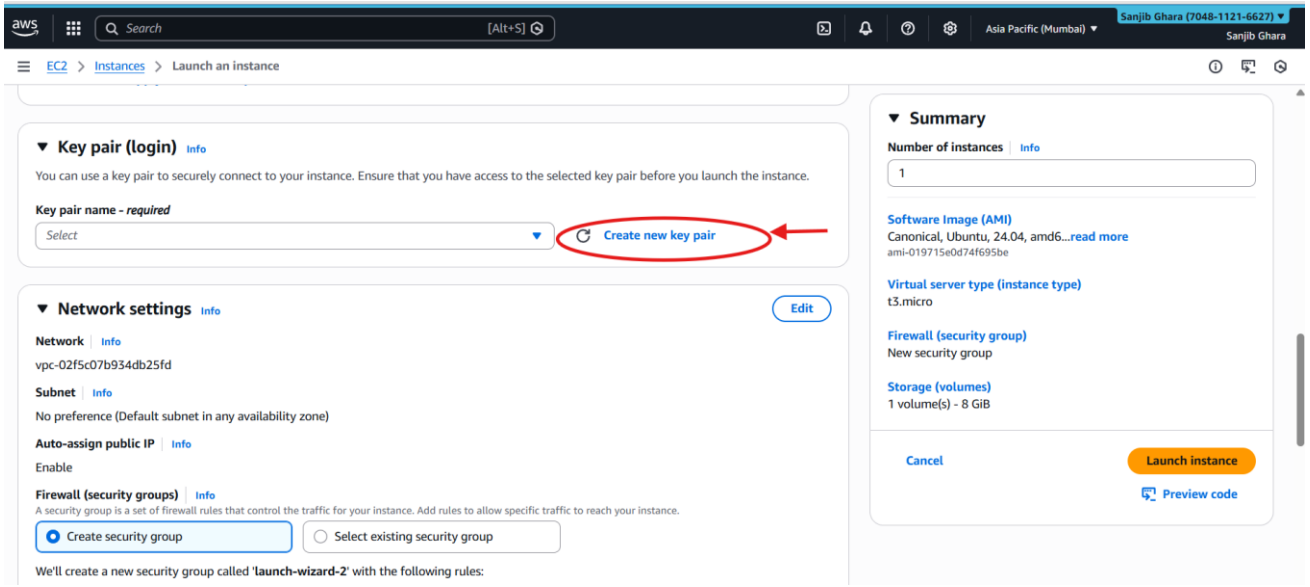
<u>Instance Configuration</u>

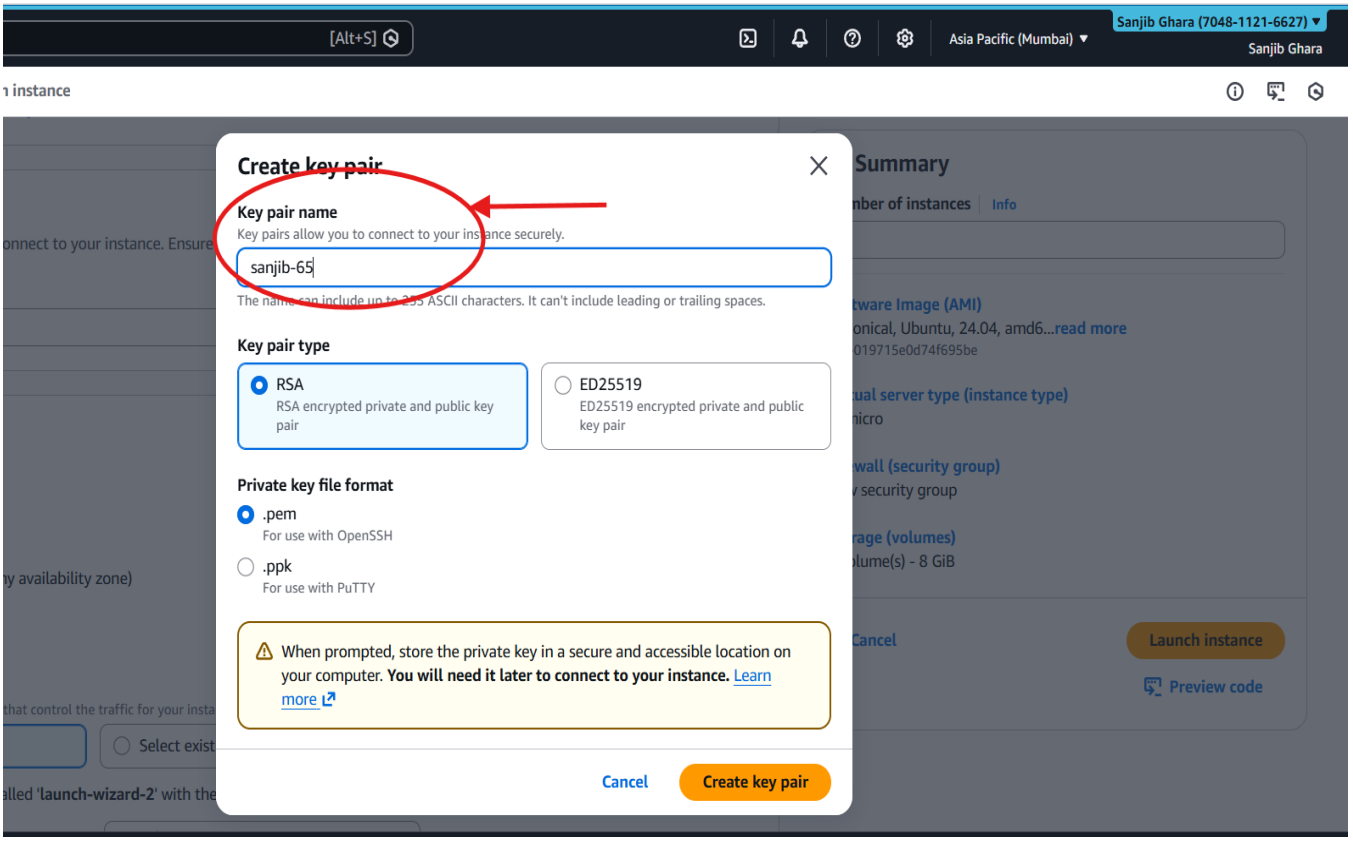
4. **Enter:**
    - Name: Sanjib-first-hosting
5. **Select:**
    - AMI: Ubuntu (Free Tier Eligible)



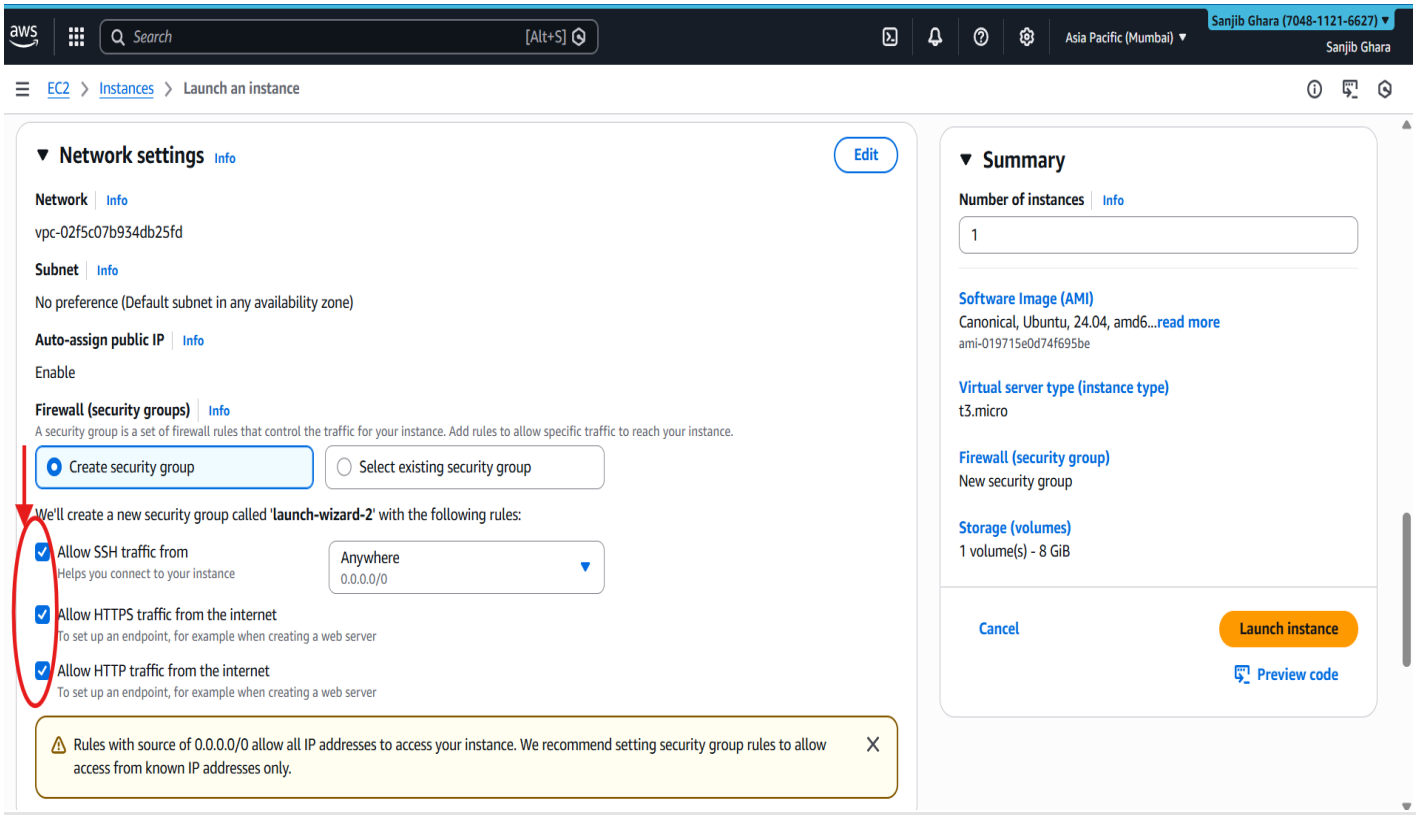


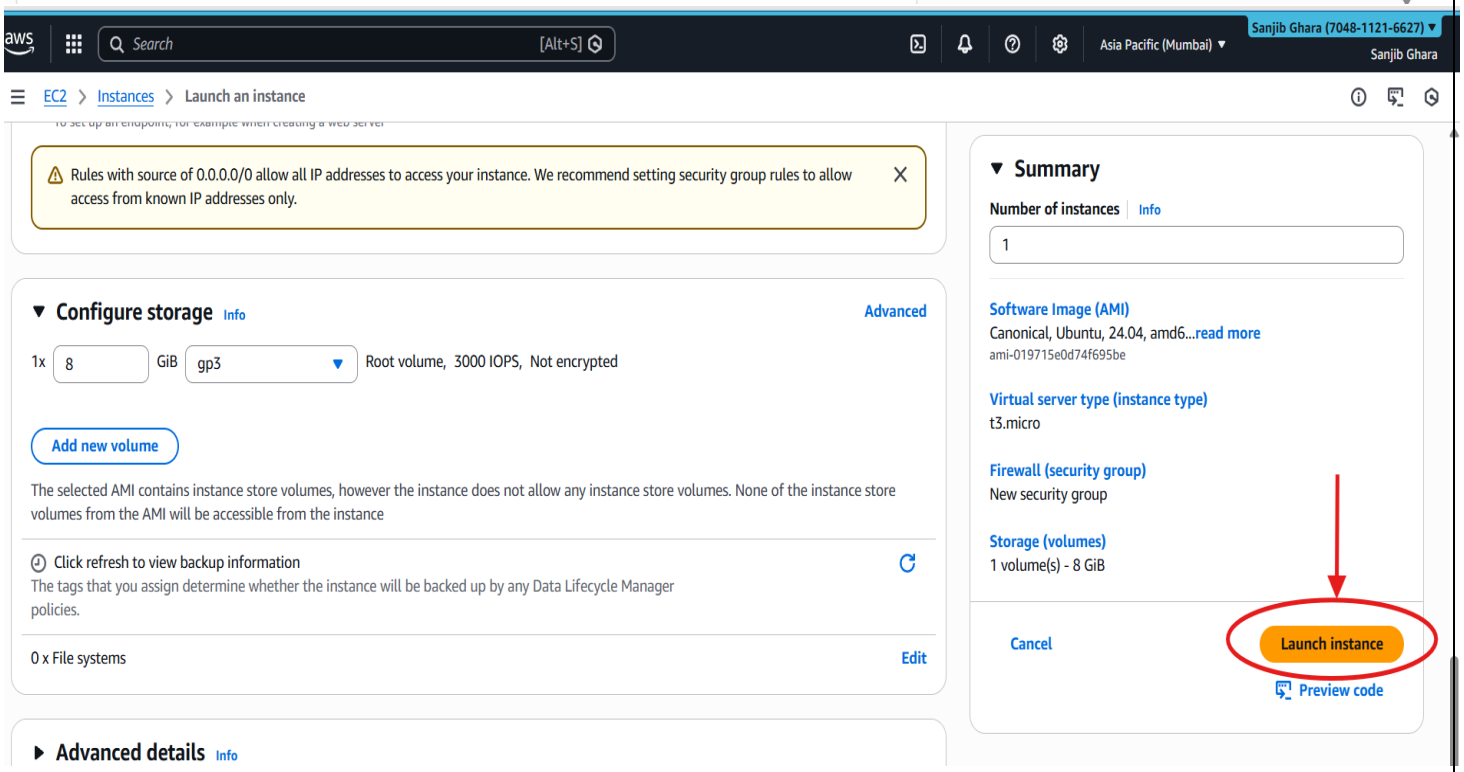
6. **Click Create new key pair:**
   - Name: Sanjib-65
   - Type: RSA
   - Format: .pem



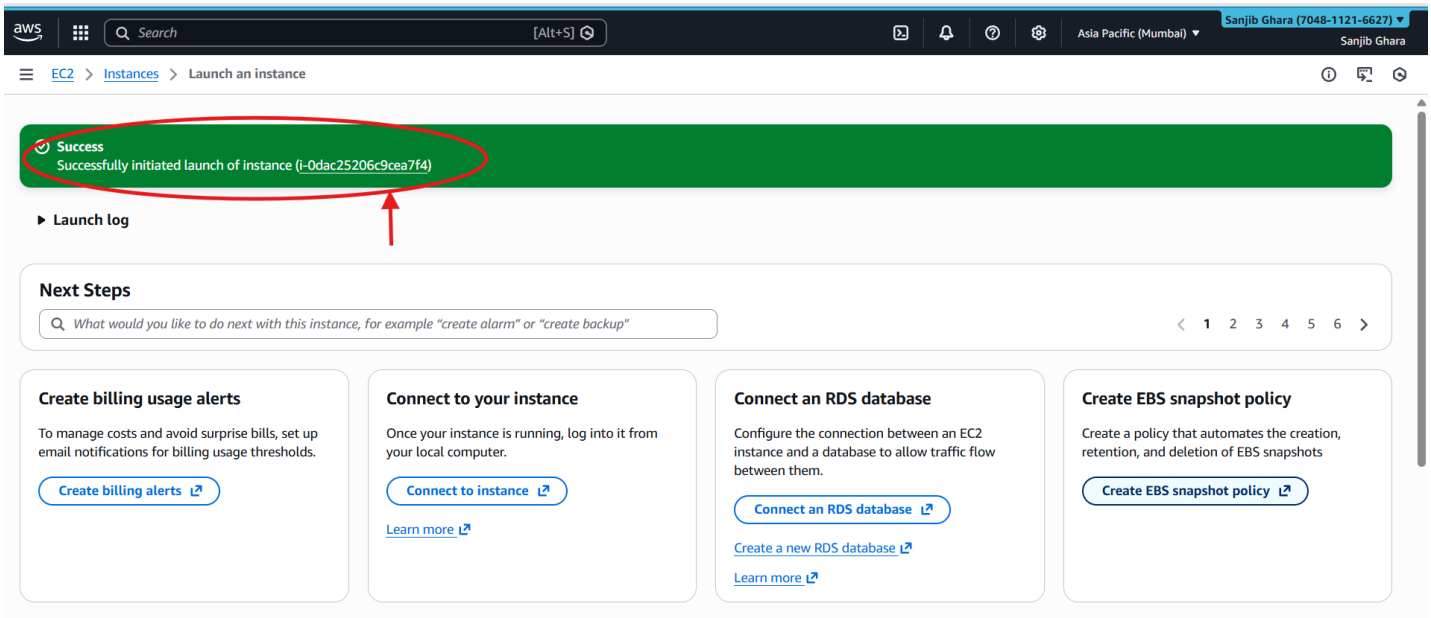


7. Download and save the key file securely.
8. Click on **Create security group** and check the boxes of **Allow HTTPS traffic from the internet and Allow HTTP traffic from the internet** and click on **Launch instance** button.

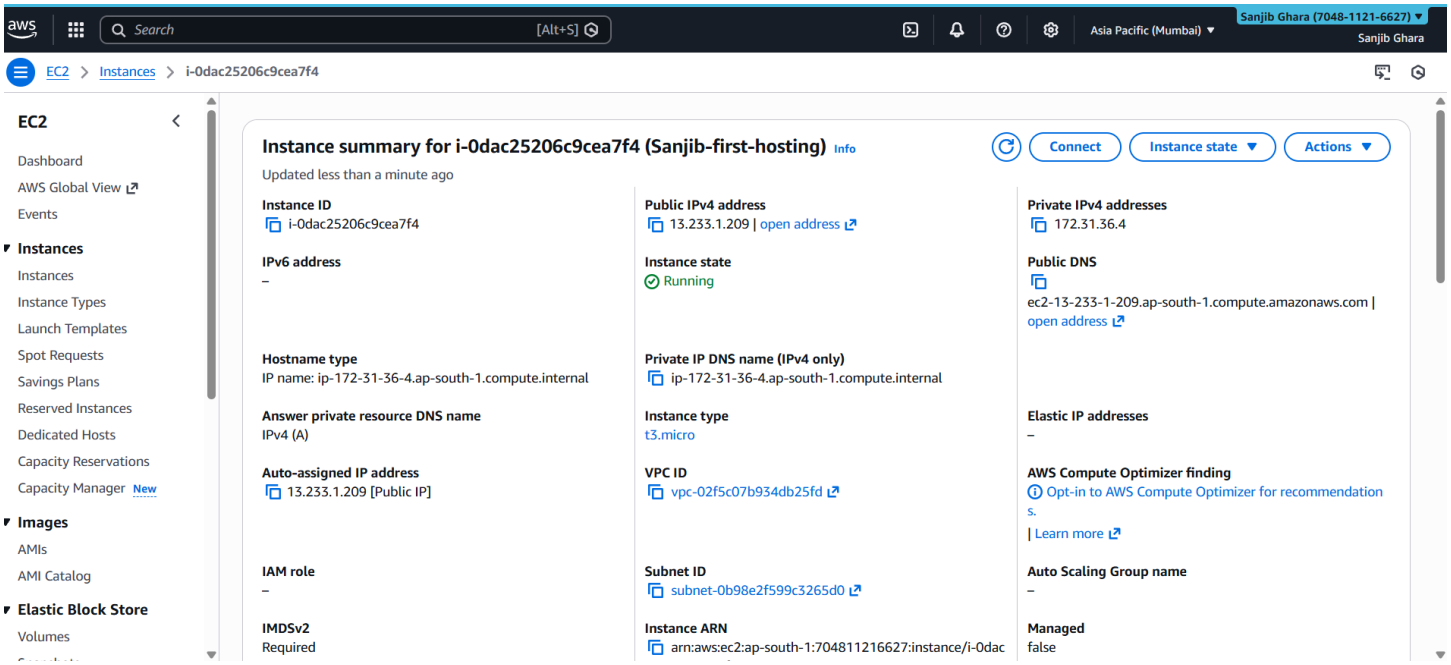




9. The Instance will be successfully created , return to the **Instance** page.



10: The Instance will be seen running, click on the newly created instance

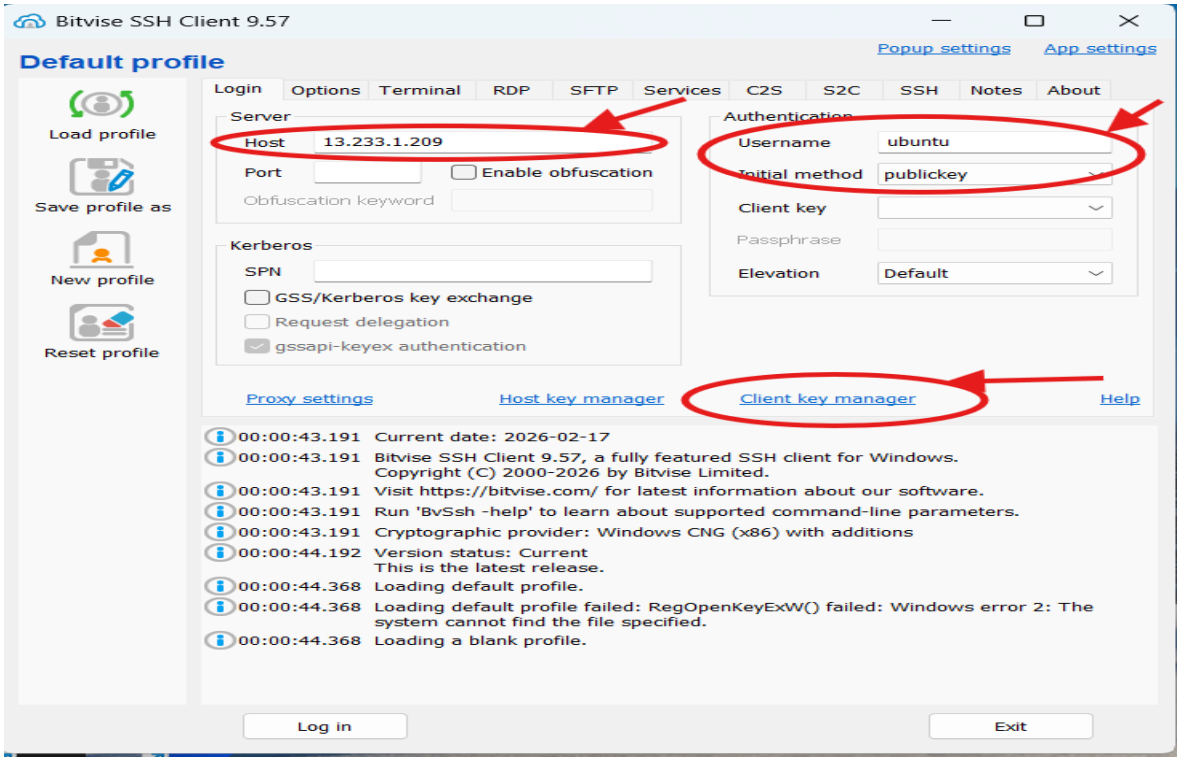


Step 2: Copy Public IPv4 Address

After the EC2 instance is launched and running, copy the Public IPv4 address from the Instance details page.

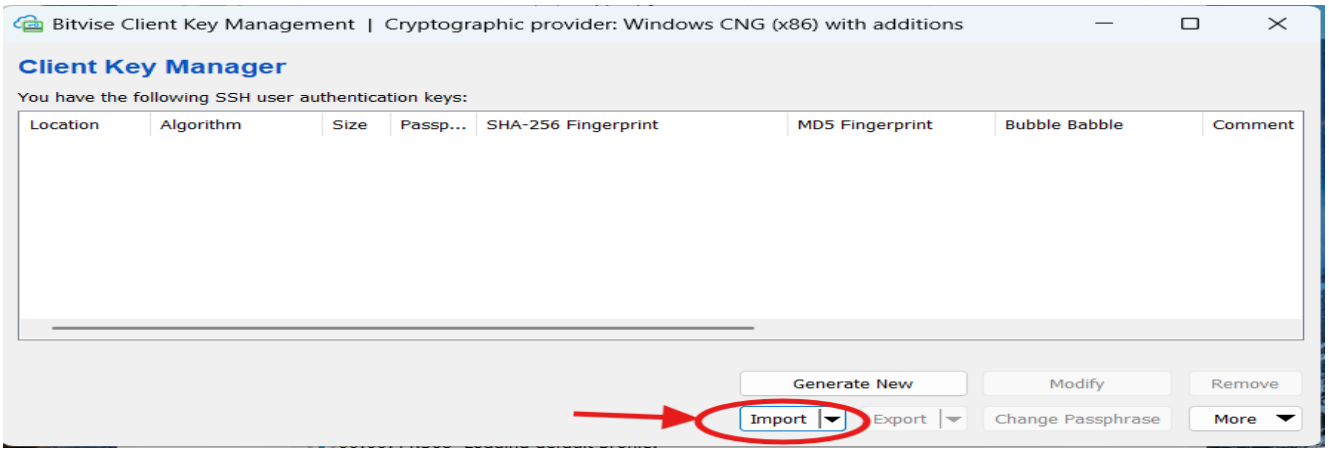
Step 3: Open Bitvise SSH Client and Configure Connection

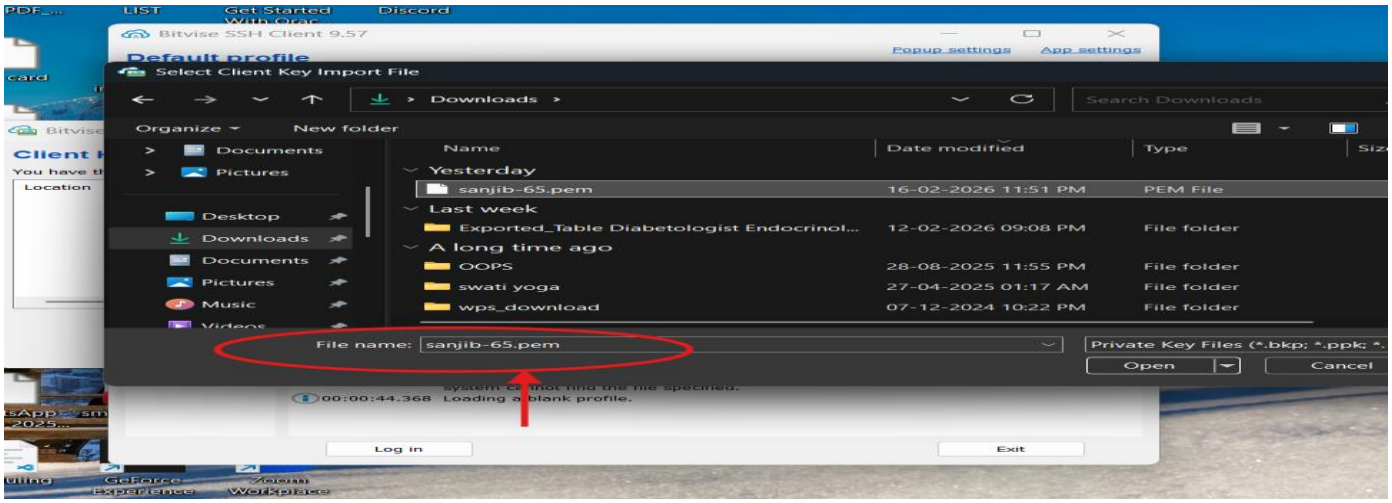
1. Open Bitvise SSH Client on the computer.
2. Paste the copied Public IPv4 address into the Host field.
3. Enter ubuntu in the Username field.
4. Set Initial method to publickey.
5. Click on Client key manager

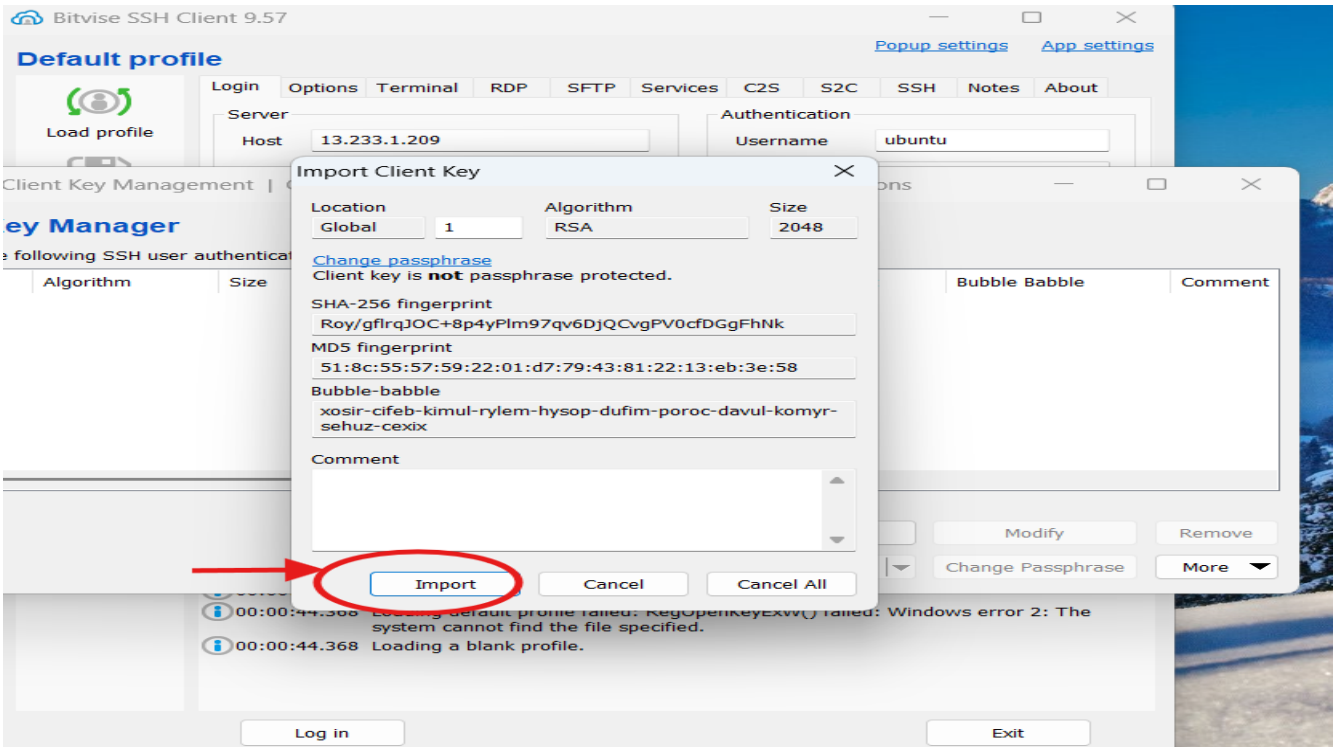


Step 4: Import Private Key

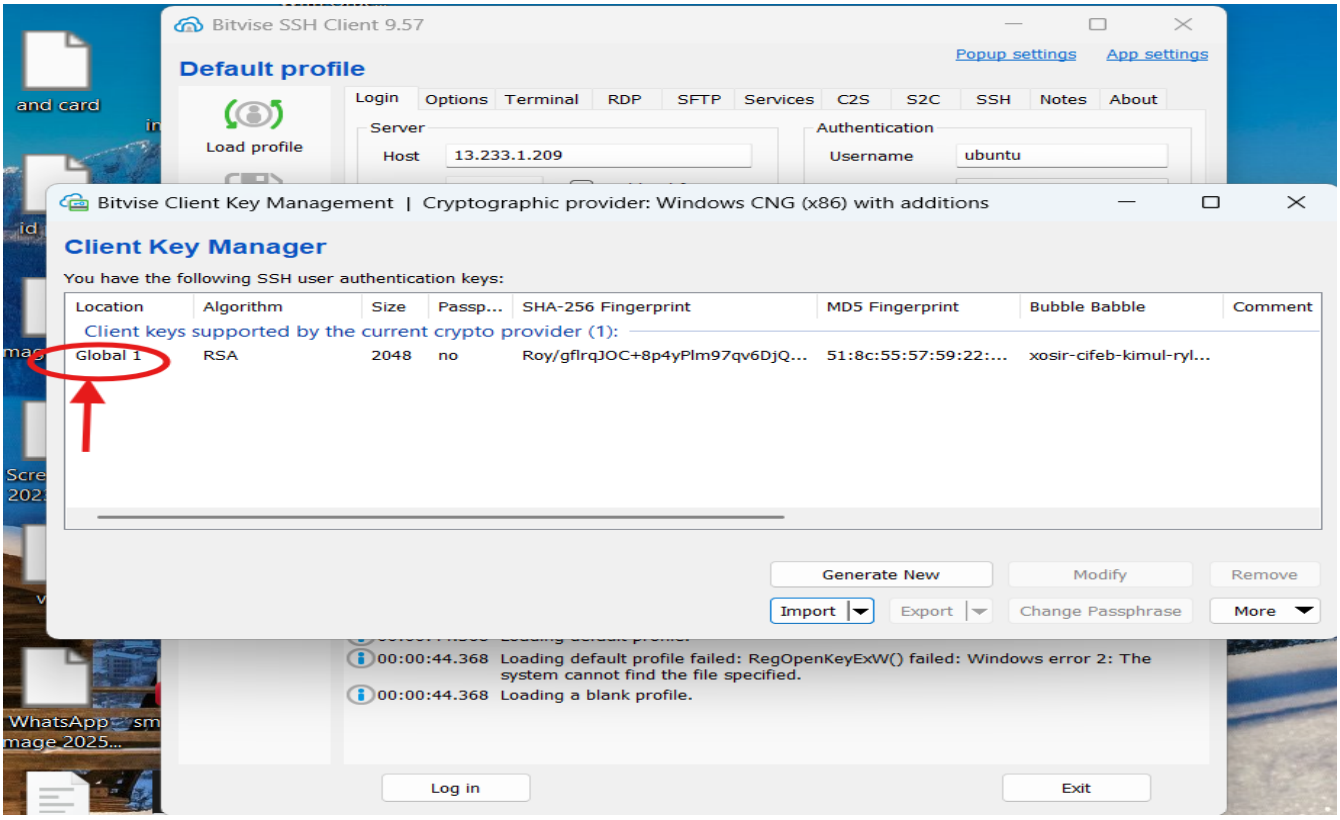
1. In Client Key Manager, click Import.
2. Select the downloaded .pem key file from the Desktop.





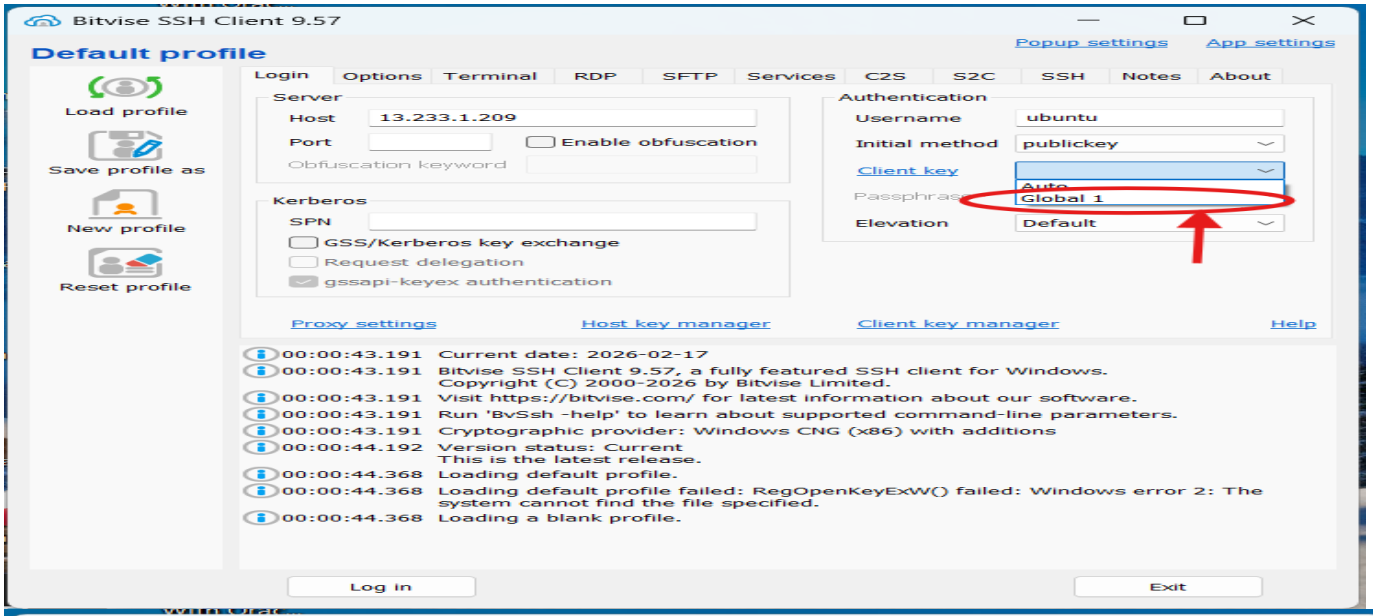


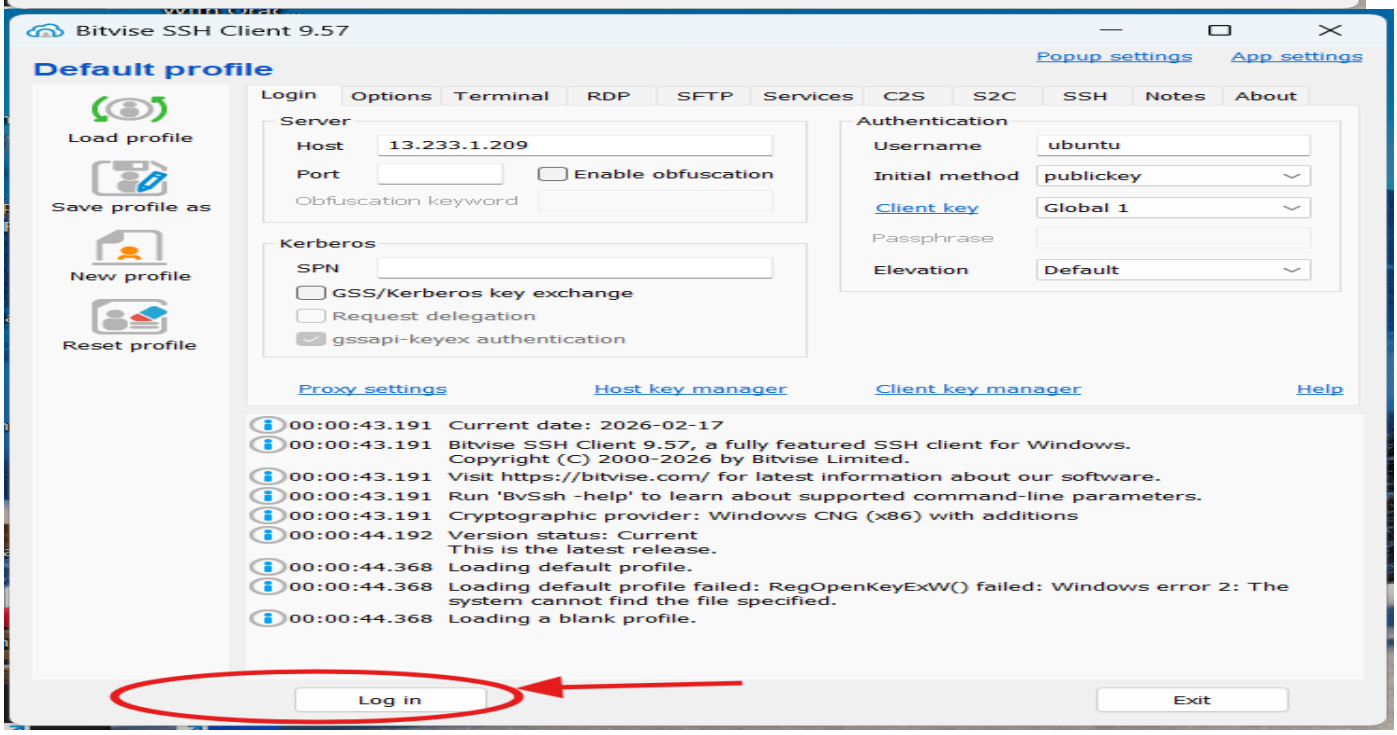
3.Save the key.



Step 5: Login to the Instance

1. Ensure Client key is set to Global 1.
2. Click on the Log in button.

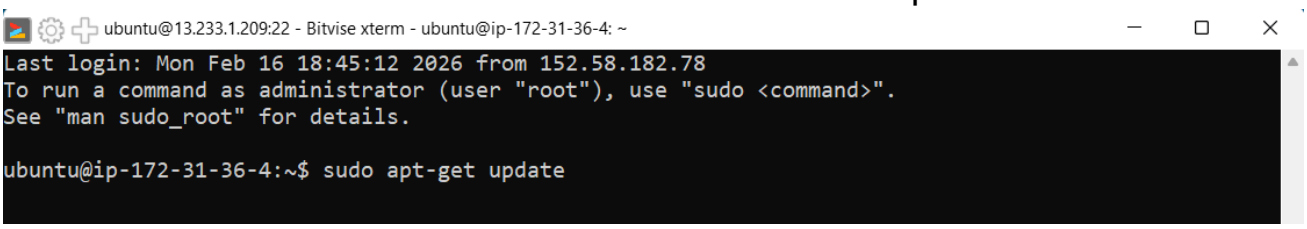




Step 6: Open Terminal Console

After successful login:

- Click on **New Terminal Console** from the **Bitvise** side panel.



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Step 7: Update Ubuntu System

Run the following command to update and upgrade the system:

```
Get:56 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [396 B]
Fetched 39.9 MB in 6s (6529 kB/s)
Reading package lists... Done
ubuntu@ip-172-31-36-4:~$ sudo apt-get upgrade
```

sudo apt-get update && sudo apt-get upgrade -y

Step 8: Install Nginx Web Server

Install Nginx using the command:

```
User sessions running outdated binaries:
 ubuntu @ session #2: sshd[1061,1172]
 ubuntu @ user manager service: systemd[1066]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-36-4:~$ sudo apt-get install nginx
```

sudo apt-get install nginx -y

```
Suggested packages:
  fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 565 kB of archives.
After this operation, 1596 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

Step 9: Set Permissions for Web Directory

Navigate to the web directory:

cd /var/www

```
ubuntu@ip-172-31-36-4:/home$ cd..
cd..: command not found
ubuntu@ip-172-31-36-4:/home$ cd ..
ubuntu@ip-172-31-36-4:/$ ls
bin                boot  etc   lib               lib64       media  opt   root  sbin               snap  sys  usr
bin.usr-is-merged  dev   home  lib.usr-is-merged lost+found  mnt    proc  run   sbin.usr-is-merged srv   tmp  var
ubuntu@ip-172-31-36-4:/$
```

```
ubuntu@ip-172-31-36-4:/$ cd var
ubuntu@ip-172-31-36-4:/var$
```

```
ubuntu@ip-172-31-36-4:/var$ ls
backups  cache  crash  lib  local  lock  log  mail  opt  run  snap  spool  tmp  www
ubuntu@ip-172-31-36-4:/var$
```

```
ubuntu@ip-172-31-36-4:/var$ cd www/
ubuntu@ip-172-31-36-4:/var/www$ ls
html
ubuntu@ip-172-31-36-4:/var/www$
```

Give proper permissions to the HTML directory:

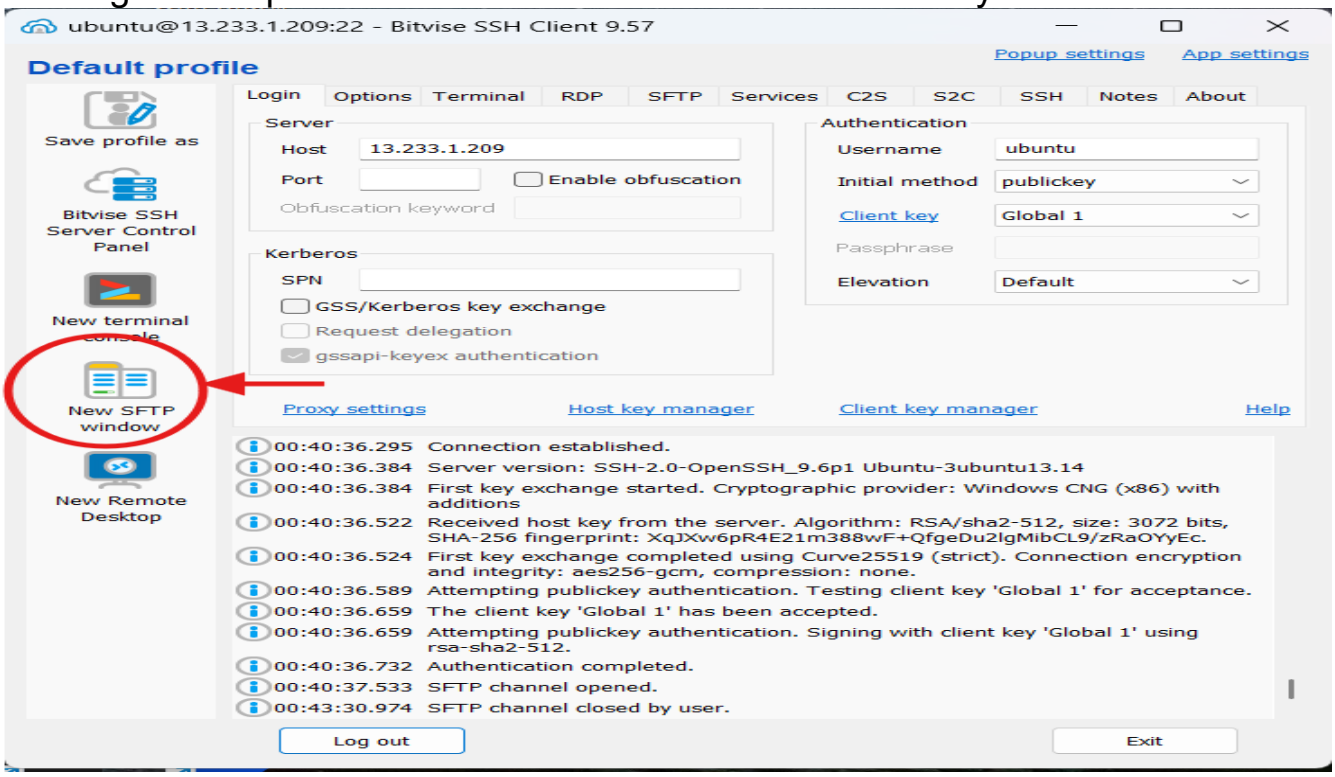
sudo chmod 777 html/

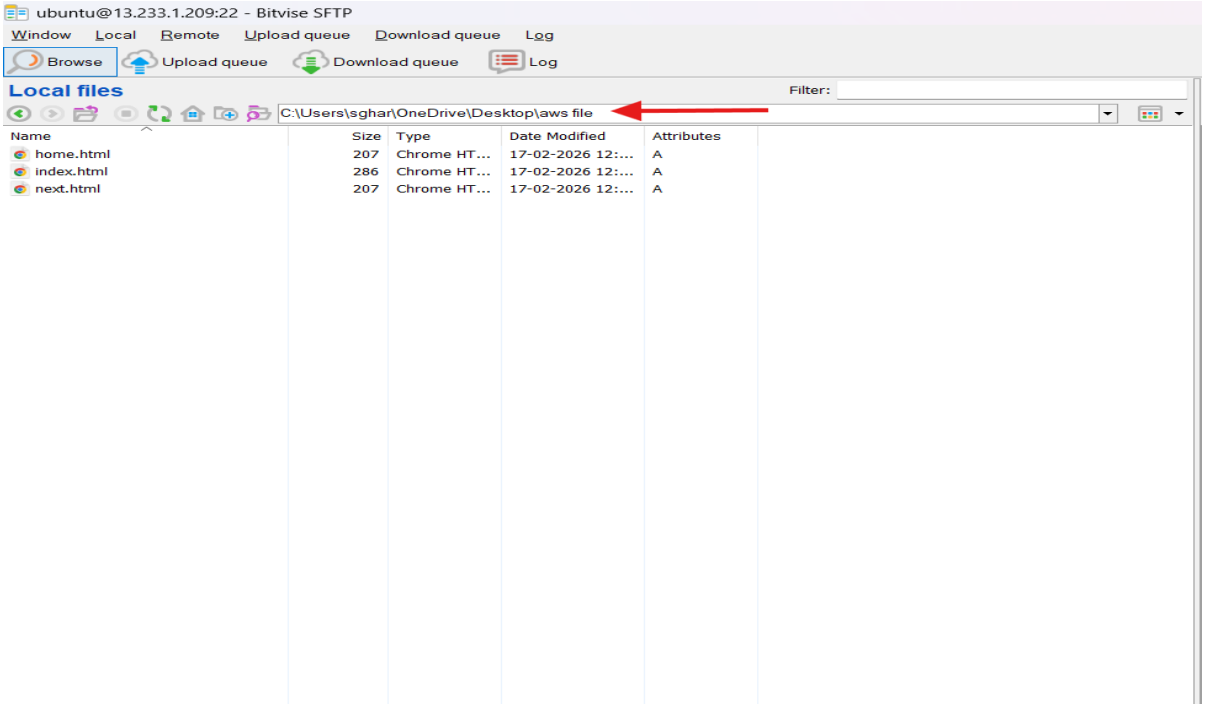
```
ubuntu@ip-172-31-36-4:/var/www$ ls
html
ubuntu@ip-172-31-36-4:/var/www$ sudo chmod 777 html
ubuntu@ip-172-31-36-4:/var/www$
```

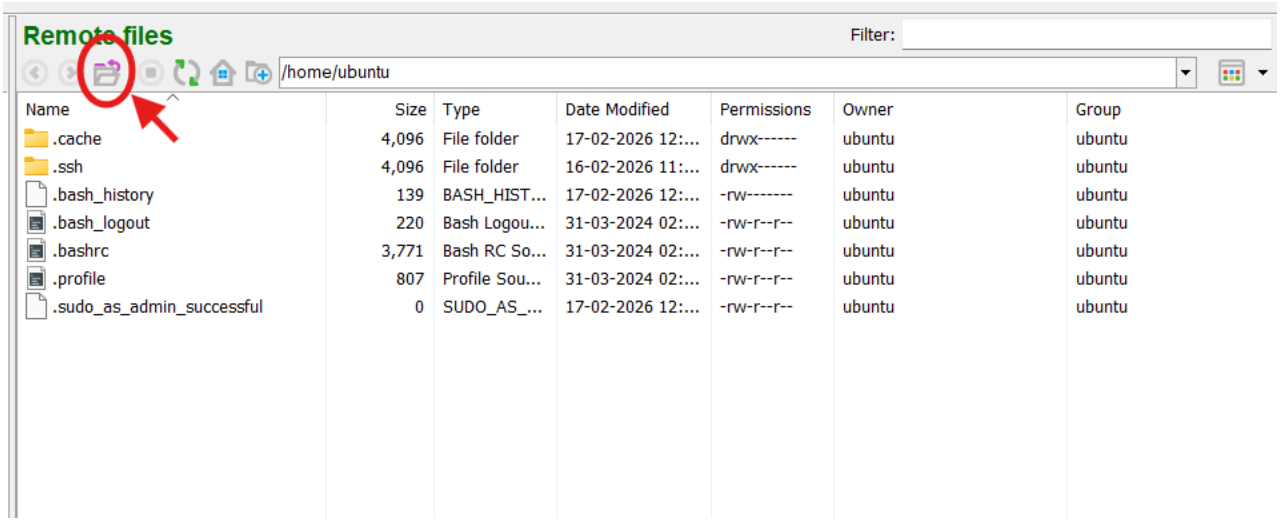
Step 10: Upload Website Files Using SFTP

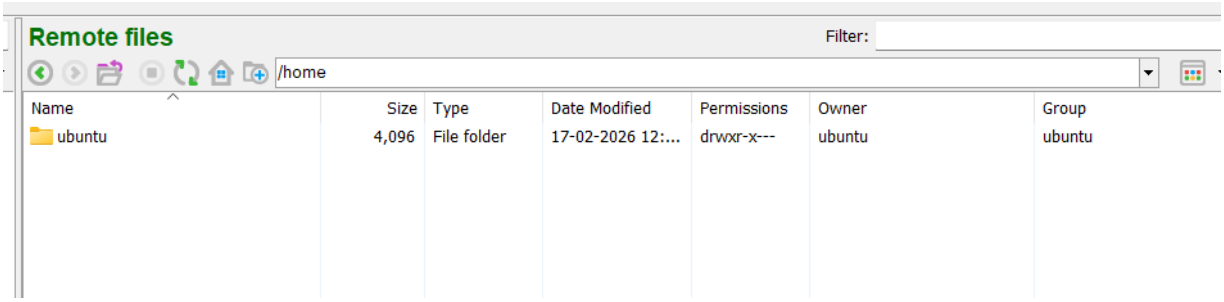
1. Click New SFTP window in Bitvise.
2. In the left pane (Local Files), navigate to the folder containing:
    - index.html
    - prev.html
    - next.html

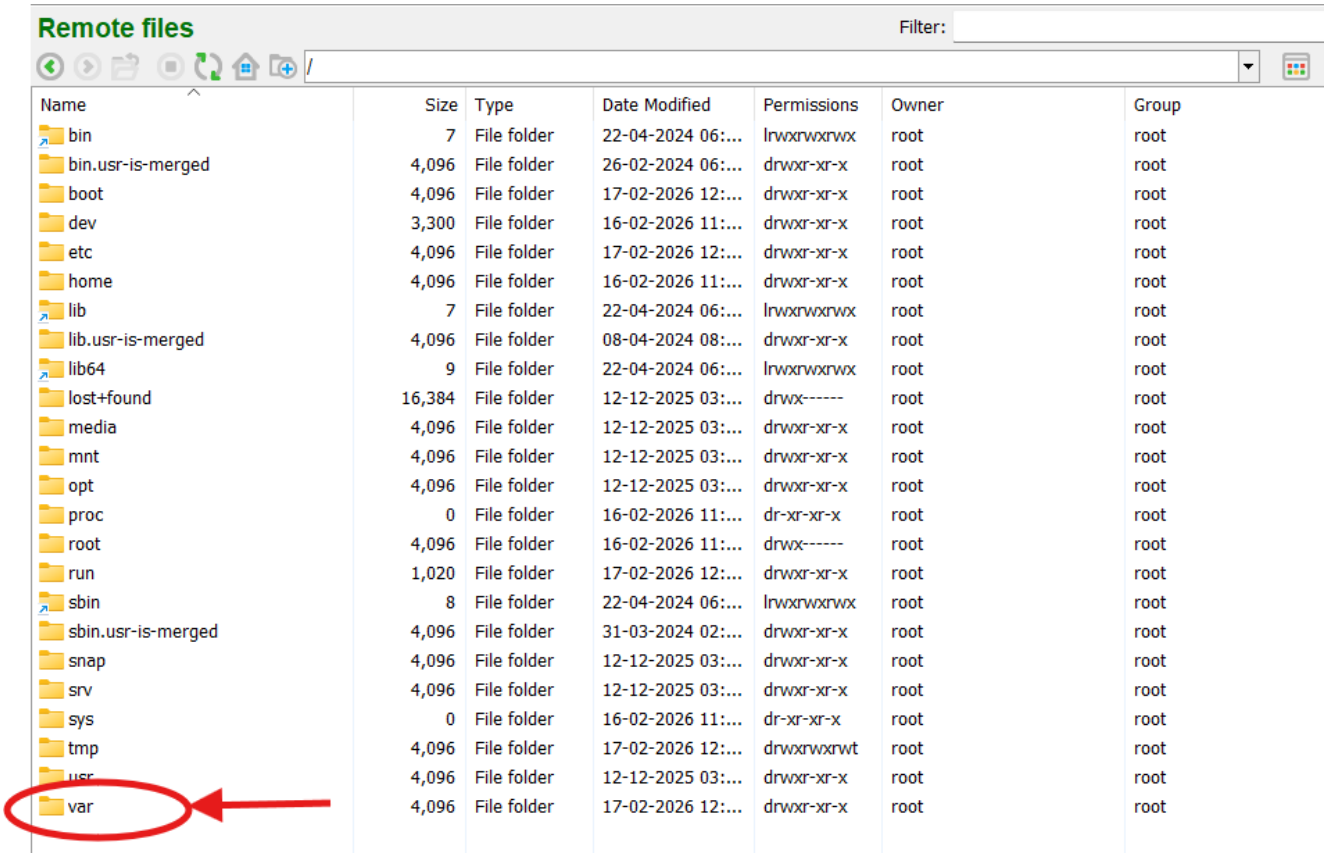
3. In the right pane (Remote Files), navigate to: **/var/www/html**
4. Drag and drop all three files from local to remote directory.

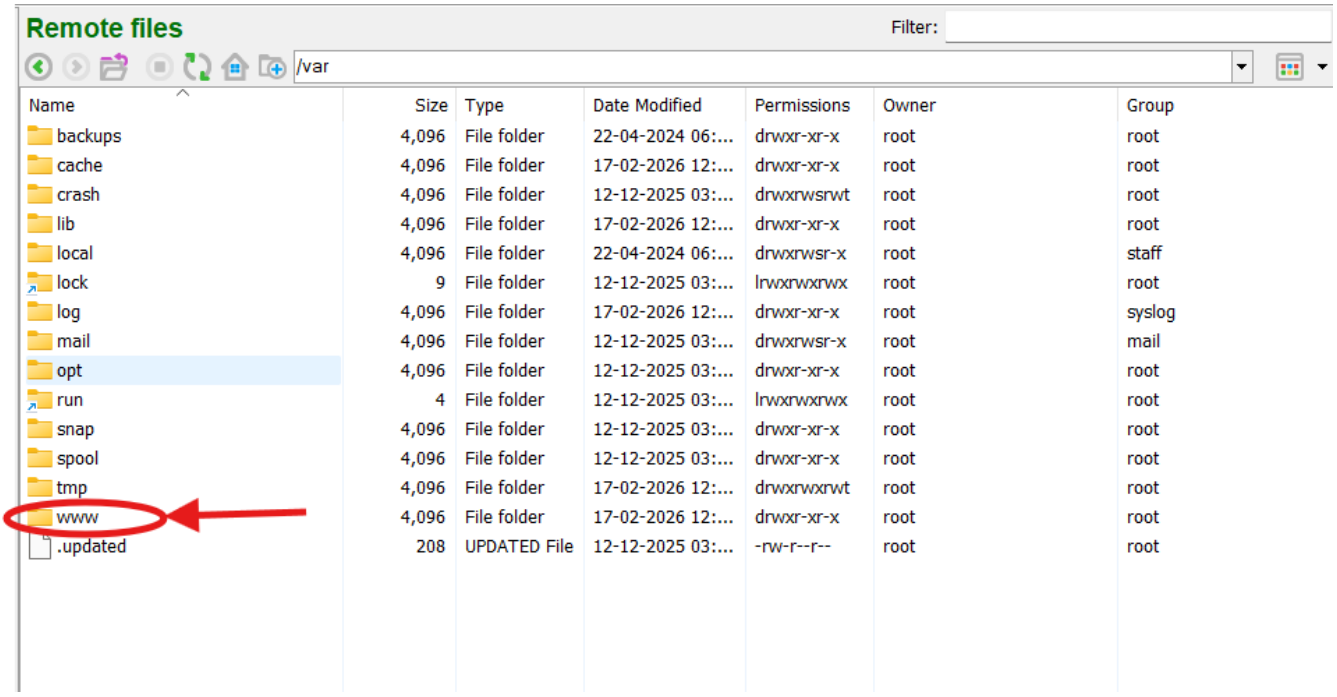


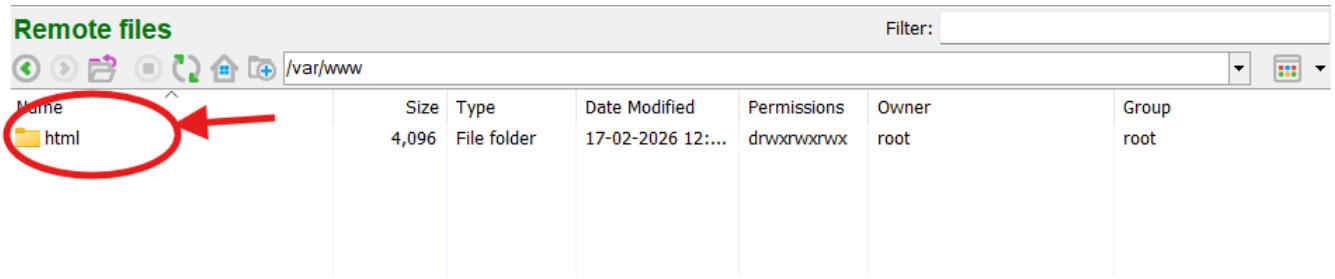


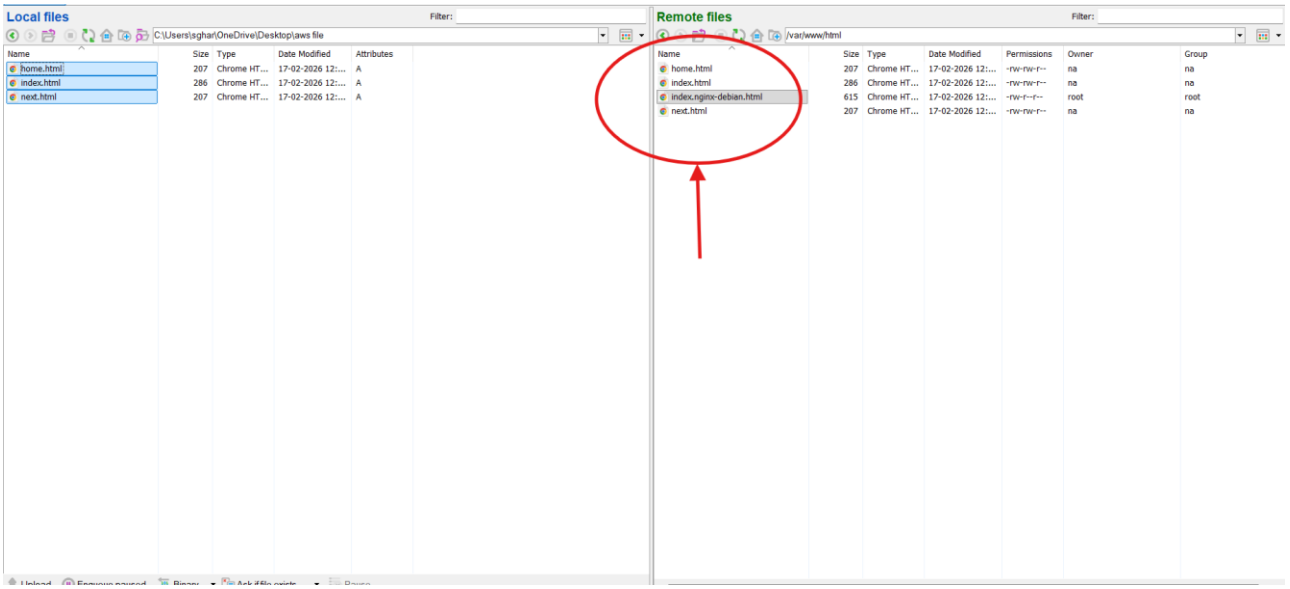

Remote files
Filter:
/home/ubuntu
Name Size Type Date Modified Permissions Owner Group
.cache 4,096 File folder 17-02-2026 12:... drwx------ ubuntu ubuntu
.ssh 4,096 File folder 16-02-2026 11:... drwx------ ubuntu ubuntu
.bash_history 139 BASH_HIST... 17-02-2026 12:... -rw------- ubuntu ubuntu
.bash_logout 220 Bash Logou... 31-03-2024 02:... -rw-r--r-- ubuntu ubuntu
.bashrc 3,771 Bash RC So... 31-03-2024 02:... -rw-r--r-- ubuntu ubuntu
.profile 807 Profile Sou... 31-03-2024 02:... -rw-r--r-- ubuntu ubuntu
.sudo_as_admin_successful 0 SUDO_AS_... 17-02-2026 12:... -rw-r--r-- ubuntu ubuntu


Remote files
Filter:
/home
Name Size Type Date Modified Permissions Owner Group
ubuntu 4,096 File folder 17-02-2026 12:... drwxr-x--- ubuntu ubuntu


Remote files
Filter:
/
Name Size Type Date Modified Permissions Owner Group
bin 7 File folder 22-04-2024 06:... lrwxrwxrwx root root
bin.usr-is-merged 4,096 File folder 26-02-2024 06:... drwxr-xr-x root root
boot 4,096 File folder 17-02-2026 12:... drwxr-xr-x root root
dev 3,300 File folder 16-02-2026 11:... drwxr-xr-x root root
etc 4,096 File folder 17-02-2026 12:... drwxr-xr-x root root
home 4,096 File folder 16-02-2026 11:... drwxr-xr-x root root
lib 7 File folder 22-04-2024 06:... lrwxrwxrwx root root
lib.usr-is-merged 4,096 File folder 08-04-2024 08:... drwxr-xr-x root root
lib64 9 File folder 22-04-2024 06:... lrwxrwxrwx root root
lost+found 16,384 File folder 12-12-2025 03:... drwx------ root root
media 4,096 File folder 12-12-2025 03:... drwxr-xr-x root root
mnt 4,096 File folder 12-12-2025 03:... drwxr-xr-x root root
opt 4,096 File folder 12-12-2025 03:... drwxr-xr-x root root
proc 0 File folder 16-02-2026 11:... dr-xr-xr-x root root
root 4,096 File folder 16-02-2026 11:... drwx------ root root
run 1,020 File folder 17-02-2026 12:... drwxr-xr-x root root
sbin 8 File folder 22-04-2024 06:... lrwxrwxrwx root root
sbin.usr-is-merged 4,096 File folder 31-03-2024 02:... drwxr-xr-x root root
snap 4,096 File folder 12-12-2025 03:... drwxr-xr-x root root
srv 4,096 File folder 12-12-2025 03:... drwxr-xr-x root root
sys 0 File folder 16-02-2026 11:... dr-xr-xr-x root root
tmp 4,096 File folder 17-02-2026 12:... drwxrwxrwt root root
usr 4,096 File folder 12-12-2025 03:... drwxr-xr-x root root
var 4,096 File folder 17-02-2026 12:... drwxr-xr-x root root


Remote files
Filter:
/var
Name Size Type Date Modified Permissions Owner Group
backups 4,096 File folder 22-04-2024 06:... drwxr-xr-x root root
cache 4,096 File folder 17-02-2026 12:... drwxr-xr-x root root
crash 4,096 File folder 12-12-2025 03:... drwxrwsrwt root root
lib 4,096 File folder 17-02-2026 12:... drwxr-xr-x root root
local 4,096 File folder 22-04-2024 06:... drwxrwsr-x root staff
lock 9 File folder 12-12-2025 03:... lrwxrwxrwx root root
log 4,096 File folder 17-02-2026 12:... drwxr-xr-x root syslog
mail 4,096 File folder 12-12-2025 03:... drwxrwsr-x root mail
opt 4,096 File folder 12-12-2025 03:... drwxr-xr-x root root
run 4 File folder 12-12-2025 03:... lrwxrwxrwx root root
snap 4,096 File folder 12-12-2025 03:... drwxr-xr-x root root
spool 4,096 File folder 12-12-2025 03:... drwxr-xr-x root root
tmp 4,096 File folder 17-02-2026 12:... drwxrwxrwt root root
www 4,096 File folder 17-02-2026 12:... drwxr-xr-x root root
.updated 208 UPDATED File 12-12-2025 03:... -rw-r--r-- root root


Remote files
Filter:
/var/www
Name Size Type Date Modified Permissions Owner Group
html 4,096 File folder 17-02-2026 12:... drwxrwxrwx root root


Local files
Remote files
home.html
index.html
index.nginx-debian.html
next.html

Step 11: Access the Website

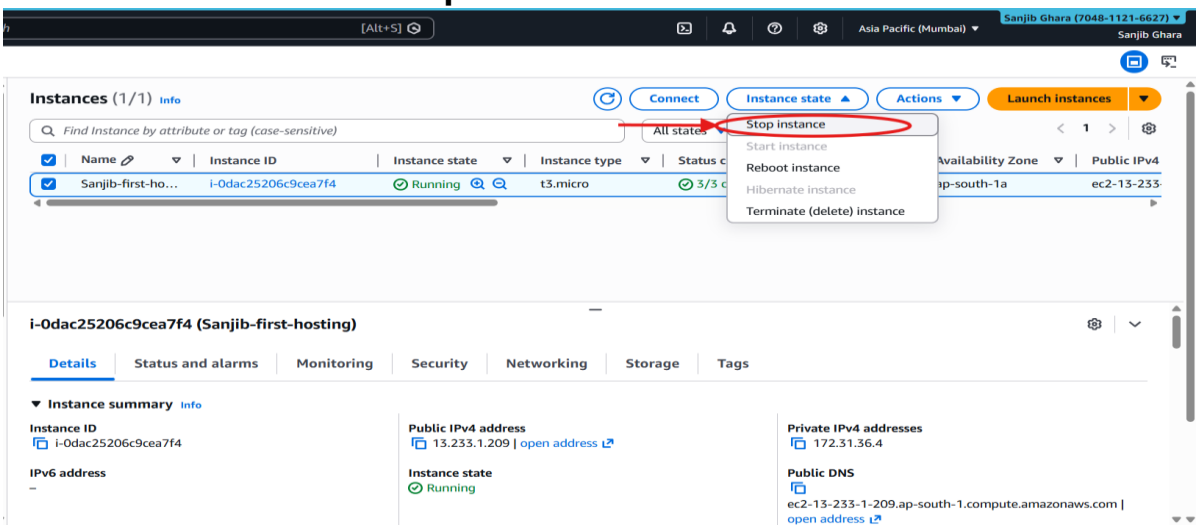
Open a new browser tab and enter:

http://<Public-IPv4-address>

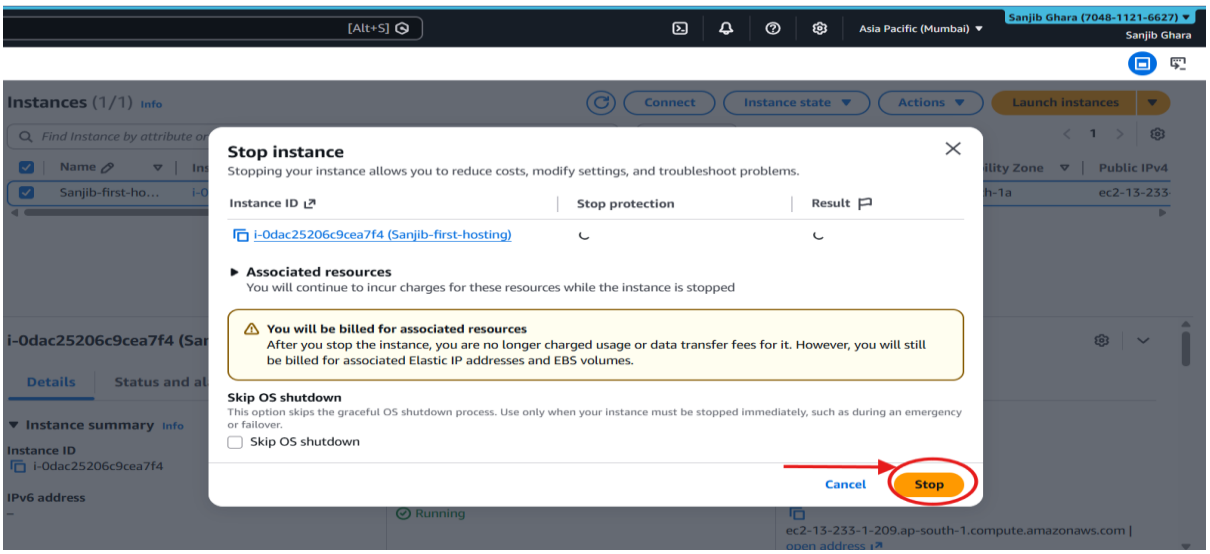
The static website will load successfully.

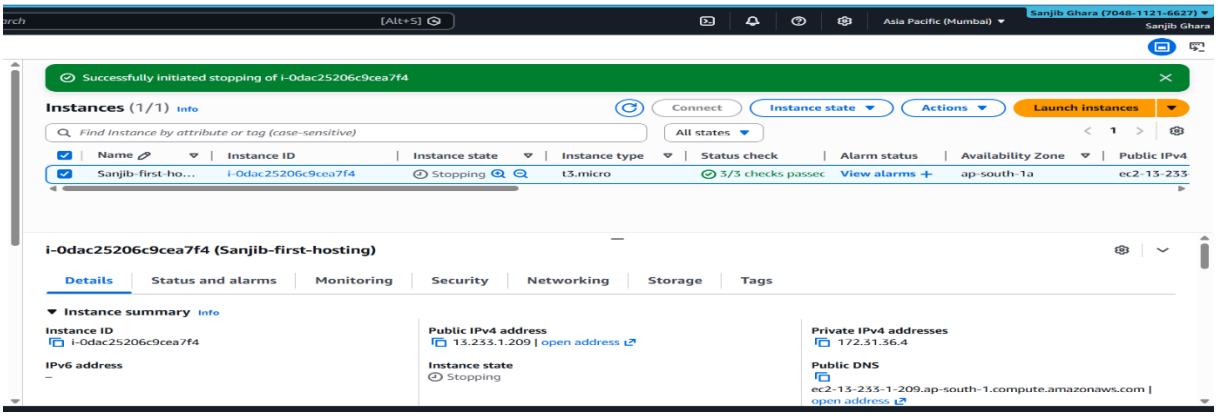
Step 12: Stop the EC2 Instance

1. Go back to the EC2 Instances page.
2. Select the instance.
3. Click Instance state → **Stop instance.**



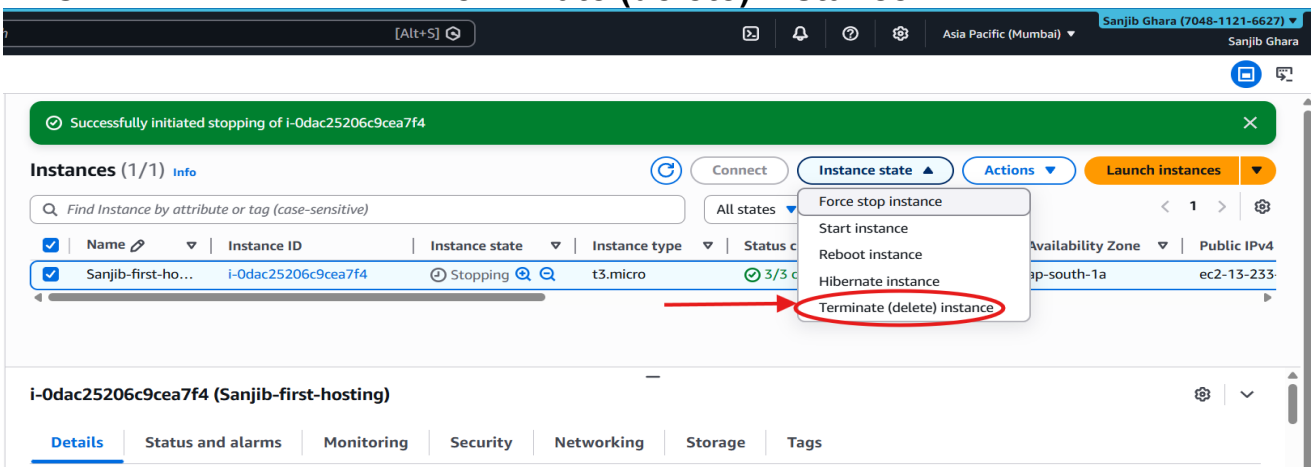
4. Confirm by clicking Stop.



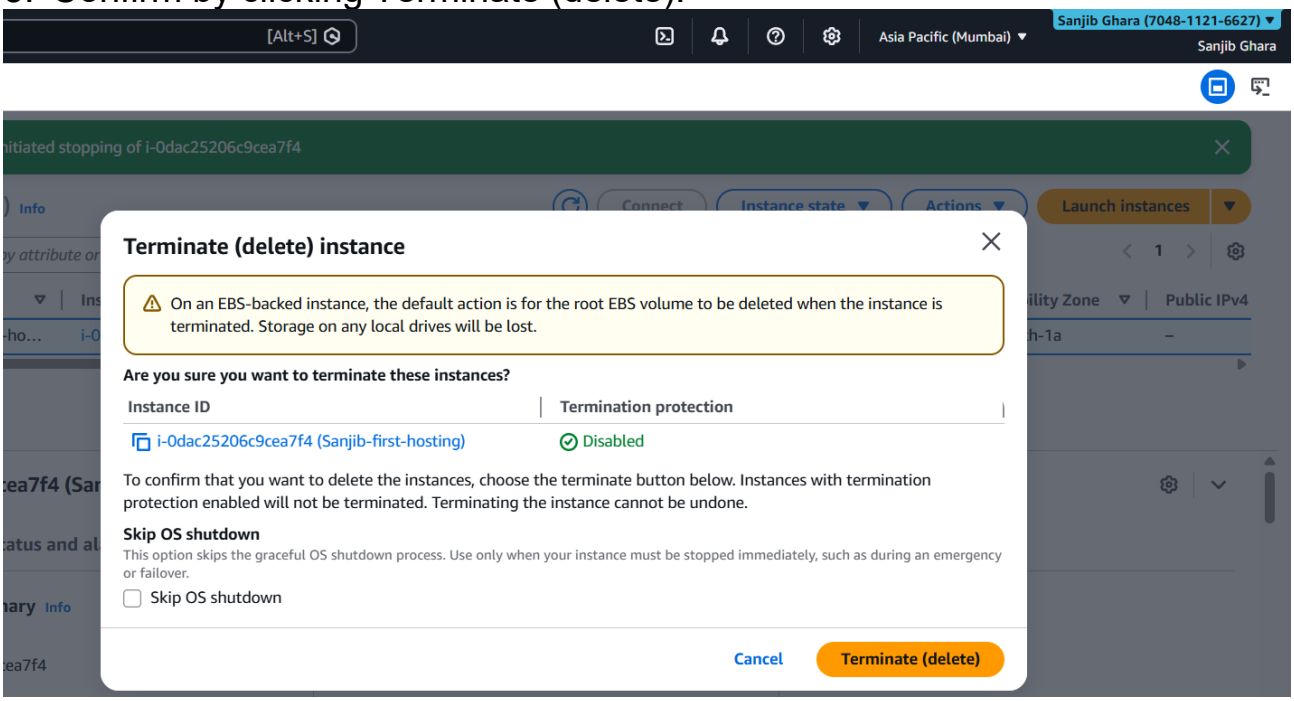


Step 13: Terminate the EC2 Instance

1. Select the same instance again.
2. Click Instance state → **Terminate (delete) instance.**

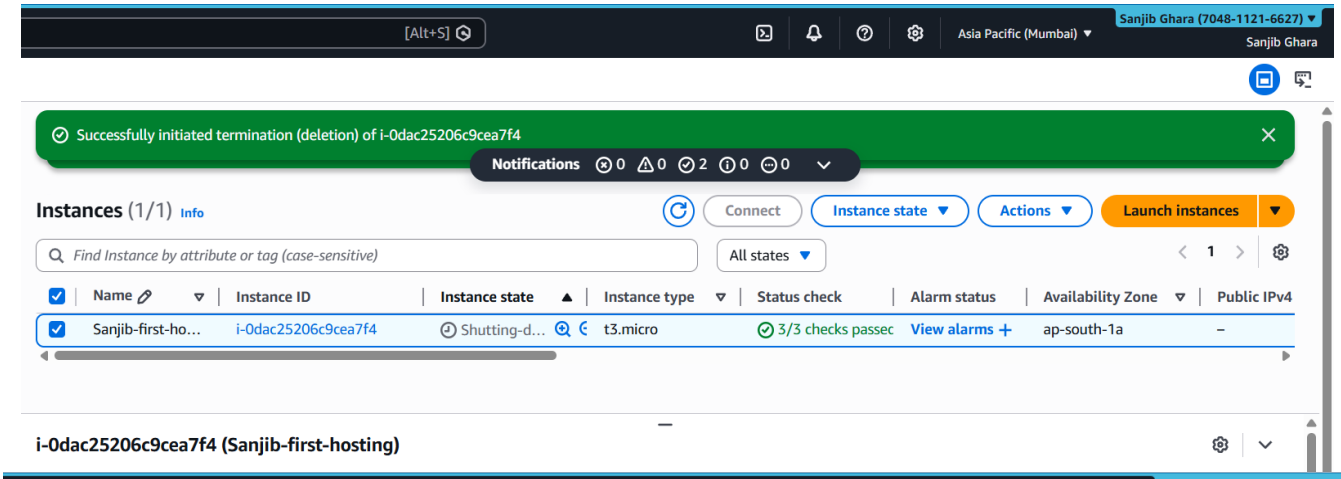


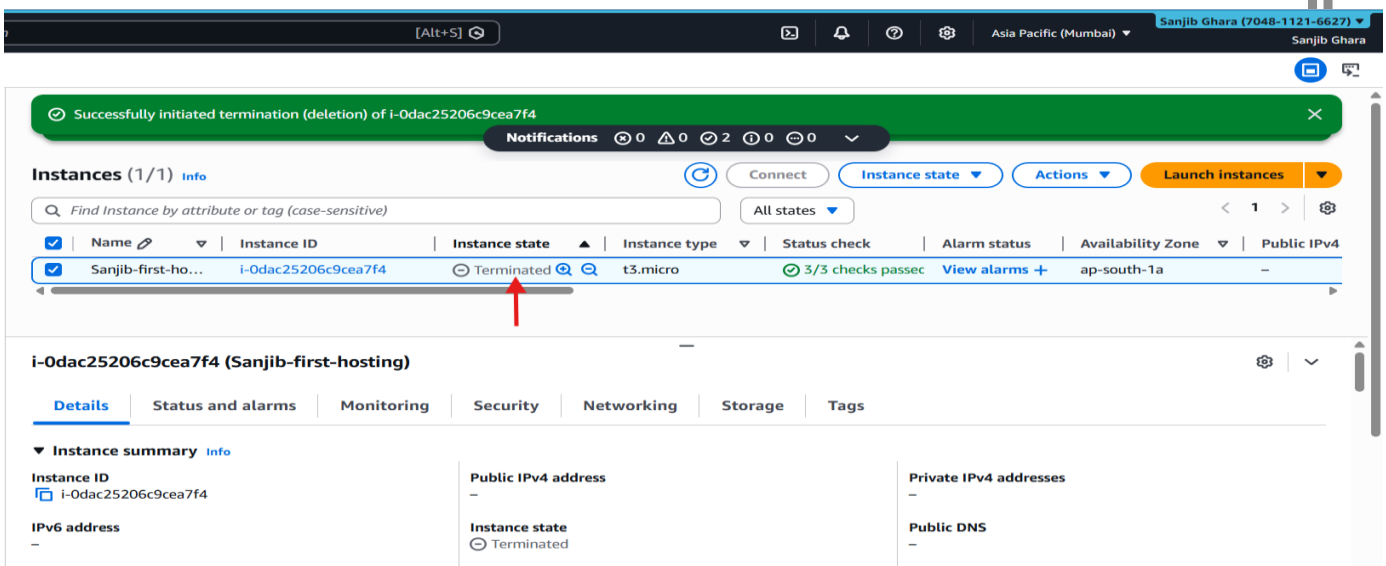
3. Confirm by clicking Terminate (delete).



Step 14: Instance Termination Confirmation

The EC2 instance is successfully terminated and all associated compute resources are released.





Conclusion

A static website consisting of multiple HTML files was successfully hosted on an Ubuntu-based EC2 instance using the Nginx web server and accessed via a public IPv4 address.